

Model selection

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Last time: Activity

- ▶ LRT will choose the full model if $G > \chi_{q,\alpha}^2$
- ▶ AIC will choose the full model if $G > 2q$
- ▶ LRT is more strict when $\chi_{q,\alpha}^2 > 2q$

```
alpha = 0.05  
q <- 1:10  
qchisq(1 - alpha, q) - 2*q
```

At $\alpha=0.05$, LRT is
more stringent for $q \leq 7$

```
## [1] 1.84145882 1.99146455 1.81472790 1.48772904 1.06343611 0.71779833 0.41145553  
## [7] 0.06714045 -0.49268694 -1.08102240 -1.69296195 -2.28136161 -2.84532635 -3.38267243
```

```
alpha = 0.01  
q <- 1:20  
qchisq(1 - alpha, q) - 2*q
```

At $\alpha=0.01$, LRT is
more stringent for $q \leq 15$

```
## [1] 4.634897e+00 5.210340e+00 5.344867e+00 5.276704e+00 5.008196e+00 4.642399e+00  
## [6] 4.811894e+00 4.475307e+00 4.090235e+00 3.665994e+00 3.214399e+00 2.744897e+00  
## [11] 2.724970e+00 2.216967e+00 1.688250e+00 1.141238e+00 6.626196e-01 2.214399e-01  
## [16] -7.309118e-05 -5.913364e-01 -1.194694e+00 -1.809133e+00 -2.382672e+00 -2.945326e+00
```

AIC vs. LRT

- ▶ LRT can only be used with nested models; AIC can compare non-nested models
- ▶ LRT is more stringent when q (difference in number of parameters between models) is small
- ▶ AIC is more stringent when q is large
- ▶ But, they ask different questions!
 - ▶ LRT: Is difference in model deviance due to chance?
 - ▶ AIC: Estimate expected prediction error
- ▶ **Neither** is asking whether difference in models is **meaningful** (*practical significance*)

Automated model search

Potential use case:

- ▶ *Many* explanatory variables
- ▶ Research question is not an inferential question about a particular variable of interest
- ▶ Want to identify a subset of potentially useful variables for predicting response

Some types:

- Best subset selection
- Stepwise selection

Other options

- PCA, other ways to reduce dimensionality
- Penalization (elastic net, ridge, lasso, etc.)

Best subset selection (BSS)

- ▶ Fit *all* possible models (for every possible combination of the explanatory variables)
- ▶ Choose the model with the best error (e.g. measured by AIC or BIC)

Advantage: No specify full model,
BSS will look at all possible
submodels

Disadvantage: may be computationally prohibitive!
(2^H possible models)

Stepwise procedures

Greedy algorithms!

Forward selection:

- begin with a small model
- sequentially add variables, choosing variable that best improves criterion (e.g. AIC or BIC)

Backward selection:

- begin with large model
- sequentially remove variables, dropping variable that best improves criterion

Bidirectional selection:

- on each step, can either add or remove a variable

Example

```
dengue_bss <- bestglm::bestglm(dengue_sub, binomial, "AIC")
```

```
## Morgan-Tatar search since family is non-gaussian.
```

```
dengue_bss$BestModel$coefficients
```

```
##      (Intercept)           Age      Vomiting      Temperature
## -10.401598512    0.127824889    0.397322146    0.271530634
##           WBC           PLT
## -0.267207869   -0.005583689
```

Example

```
# forward selection, beginning with empty model
m_null <- glm(Dengue ~ 1, data = dengue_sub, family = binom)
m_forward <- MASS::stepAIC(m_null, scope = ~ Sex + Age +
                           Vomiting + Abdominal + Temperature +
                           BMI + WBC + HCT + PLT,
                           direction = "forward",
                           trace = 0)

m_forward$coefficients
```

##	(Intercept)	WBC	Age	PLT
##	-10.401598512	-0.267207869	0.127824889	-0.005583689
##	Temperature	Vomiting		
##	0.271530634	0.397322146		

Example

```
# backward selection, beginning with full model
m_full <- glm(Dengue ~ Sex + Age + Vomiting + Abdominal +
              Temperature + BMI + WBC + HCT + PLT,
              data = dengue_sub, family = binomial)
m_backward <- MASS::stepAIC(m_full,
                            direction = "backward",
                            trace = 0)

m_backward$coefficients
```

```
##      (Intercept)                Age      Vomiting      Temperature
## -10.401598512      0.127824889      0.397322146      0.271530634
##                WBC                PLT
## -0.267207869      -0.005583689
```

Example

Simulate data with $n = 100$ observations and 15 explanatory variables, but no population relationship between $\mathbf{x}$ and Y .

Backward selection (AIC):

## (Intercept)		X2		X4		X8	
##	-0.1537789	-0.6867152		0.6188110		0.4246714	0.4416
##		X13					
##	0.4764864						

Forward selection (AIC):

## (Intercept)		X2		X4		X10	
##	-0.1579064	-0.6625417		0.6366640		-0.2757631	0.4624
##		X9		X11			
##	0.4082716	0.4037577					

Example

Forward selection (BIC):

```
## (Intercept)          X2  
##    0.1452519   -0.5541636
```

Key points

- ▶ Search methods are trying to find some set of explanatory variables that does reasonably well at predicting the response
- ▶ BSS considers all possible models (may be computationally expensive!)
- ▶ Stepwise procedures are greedy alternatives
- ▶ Different stepwise procedures don't always agree
- ▶ Using BIC tends to give a smaller model than AIC

Activity

Work on the activity (handout), then we will discuss.

Activity

```
# start with only an intercept
initial_fit <- glm(y ~ 1, data = train_data, family = binom)

# add predictors one at a time based on AIC
selected_fit <- step(initial_fit,
                      scope = formula(glm(y ~ .,
                                           data = train_data,
                                           family = binomial)),
                      direction = "forward", trace = 0,
                      steps = 25)

# 10-fold cross validation estimate of prediction error
k <- 10
boot::cv.glm(train_data, selected_fit,
              cost = logistic_deviance, # loss function to
              K = k)$delta[1] * k/nrow(train_data) # average
```

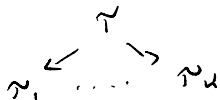
Cross-validation and model selection

CV is an estimate of R , not of $R_T(\hat{h})$

Crucial: We must cross-validate the whole model selection process, not just the final chosen model!

(model selection is part of fitting process!)

Right way:



Heldout : γ_1

Model selection : $\gamma_2, \dots, \gamma_H \rightarrow \hat{h}_{-1}$

Heldout : γ_2

Model selection : $\gamma_1, \gamma_3, \dots, \gamma_H \rightarrow \hat{h}_{-2}$

$\vdots$

Heldout : γ_H

Model selection : $\gamma_1, \dots, \gamma_{H-1} \rightarrow \hat{h}_{-H}$

the variables
chosen for
each $\hat{h}_{-i}$

may be different!

Activity

Work on the activity (course website).